

KuantorFlow — Weekly Development Report

Period: 9–13 July 2026 · **Repositories:** kuantorflow, ai_agent, kuantorflow_automation

Executive Summary

In five days, KuantorFlow grew from an empty repository into a deployed, AI-powered language-learning platform: **65 merged pull requests** across three repositories, live on PythonAnywhere.

The product today: a Flask web application where Ukrainian- and Russian-speaking learners build bilingual flashcard collections from dictionary lookups and OneNote exports, drill them with quizzes — accompanied by **Mykola**, a persona-rich AI study companion (Claude-powered, RAG-grounded) who answers questions, adds flashcards to the database on request, remembers signed-in users between visits, and greets them with a recap of their previous conversations.

Repository	Role	Merged PRs
kuantorflow	The website: flashcards, quizzes, auth, Mykola integration	35
ai_agent	The agent: RAG core, persona, tools, standalone demo app	28
kuantorflow_automation	Quality & ops: regression tests, DB backups	2
Total		65

Completed Work by Theme

1. Product foundation (kuantorflow #1–#12)

The core learning product was built in the first two days:

- **Flashcards end-to-end** (#1): parsing word lookups and OneNote .mht exports into cards, saving to MySQL, browsing by topic, and a bilingual (UA/RU) quiz mode.
- **Smarter lookups** (#2, #7): one card per part of speech, English definitions, translation gap-filling between Ukrainian and Russian; the lookup backend was switched from Reverso scraping to Google Translate's dictionary data for reliability.
- **Review before save** (#12): looked-up cards go through an edit/delete review popup instead of saving blind — a pattern later reused for file imports.
- **Polish and access control**: blue/yellow theme with topic chips (#3), card deletion with confirmation (#9), Open Graph link previews (#8), favicon (#10), environment-based DB credentials (#5), and a keyword access gate for the whole site (#11).

2. Birth of the agent (ai_agent #1–#9)

- **RAG core (#1)**: a retrieval-augmented agent answering English-learning questions from a markdown knowledge base (TF-IDF retrieval over heading-split chunks) with source citations.
- **Standalone web app (#2)**: Flask chat UI with gallery/About pages.
- **The keystone architecture decision (#9)**: the chatbot was extracted into an importable TynnaAgent class so the main site could *import* the agent rather than duplicate it — the single-source, two-repo composition that every later feature builds on.
- Early capabilities: flashcards-database access and study-list generation (#15, #16), friendly API-error messaging (#5), chat-output markdown fixes (#6, #18), a music player for the standalone app (#13, #14).

3. Integration era (kuantorflow #26–#39)

The agent moved into the main site as a floating chat widget:

- **Widget integration (#26)** with minimize/maximize, avatar animation, and an About page — served by a **cross-repo Jinja loader** that renders the shared About partial and media straight from the ai_agent checkout (no copies).
- **Two-step welcome popup** introducing the companion (#33, #37), wave-animation launcher (#39), full-page chat opened from the widget (#35), and chat logging (#32; ai_agent #24).

4. Platform hardening

- **The dependency saga** (kuantorflow #28; ai_agent #21, #22): a 1xm1 build failure on Python 3.14 (no cp314 wheel for the pinned version) was diagnosed and fixed; all dependencies were pinned, a fully-resolved requirements.lock introduced for reproducible deploys, and both repos aligned to the same production versions. Production was migrated to Python 3.13 on PythonAnywhere.
- **Request-size caps** (kuantorflow #29; ai_agent #22): 1 MB guards on chat endpoints against memory abuse and runaway API costs — scoped per-endpoint so legitimate large .mht uploads still work.
- **Log privacy** (kuantorflow #57; ai_agent #42): gitignore hardening so per-user chat logs can never be accidentally committed.

5. Identity & personalization (kuantorflow #41, #43; ai_agent #29)

- **Optional "Sign in with Google"** (OAuth 2.0 / OpenID Connect via Authlib) with an accept-or-decline consent popup — anonymous use remains fully supported, and the identity lives only in the session cookie, never in the database.
- Sign-in persists **30 days** across browser restarts; Mykola greets by full name and addresses the user by **first name** in conversation (with prompt-injection hardening on the name).

6. The Mykola transformation (ai_agent #27, #32, #34, #36, #41; kuantorflow #47)

The agent's persona evolved from Tynna to **Mykola** — a distinguished gentleman named in honour of composer Mykola Leontovych ("Shchedryk" / "Carol of the Bells"):

- Full cross-repo rename: code, routes, templates, CSS, assets, log directories.
- **A voice guide, not just a bio** (#36): British lexicon and courtesy patterns, few-shot example exchanges, moderation rules (one flourish per reply, clarity first), plus three new knowledge-base documents (French–English connections, British English, music-based learning) so the persona's signature moves are *retrieved and cited*, not improvised.

- Symbolic birthday with dynamic age (#41), copyright-aware musical style (titles, themes, short fragments — never full lyrics).

7. Agent capabilities: acting, remembering, staying current

- **Tool use** (ai_agent #37; kuantorflow #48): a real add_flashcard tool — when asked, Mykola fills all nine card fields himself (POS, definition, UA + RU translations and examples, topic) and saves through the same save_flashcard mechanism as the site's own flows, injected by kuantorflow (dependency direction preserved).
- **Per-user memory** (kuantorflow #52; ai_agent #38): signed-in users' chats are logged to mykola_logs/<email_prefix>/; on return, Mykola opens with a recap of previous conversations plus suggested follow-ups.
- **Focused context** (kuantorflow #59; ai_agent #44): recaps read only the **3 newest logs** (12k-char secondary budget), lead with the most recent topic — and if the learner already said goodbye today, Mykola deterministically wishes them a good rest instead of restarting the dialogue.
- **Readable logs** (kuantorflow #58; ai_agent #43): log files named by chat start time (chat_2026-07-13_11-43-03_e6f3.txt).
- **Live UI updates** (kuantorflow #60): when Mykola saves a card mid-chat, the "Browse flashcards" topic chips refresh in place — no reload.
- **Widget-state persistence** (kuantorflow #64; ai_agent #45, #46): the chat survives internal page navigation, keeping its history and open/maximized state.

8. Content workflow & visual identity

- **MHT two-column review** (kuantorflow #44): file uploads no longer auto-save — a popup shows the source text beside editable parsed cards with per-card delete and an "Add All" button.
- **Imagery refresh** (kuantorflow #50, #51): new icon, main artwork, and backgrounds for the Mykola era.

9. Quality & operations (kuantorflow_automation #1, #3)

- **pytest suite** (#1): app regression tests plus live-site smoke tests.
 - **Daily database backups** (#3): scheduled dump script for the production MySQL.
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Technical Highlights

- **Two-repo composition without duplication.** The site imports the agent class, renders the agent repo's templates through a ChoiceLoader, and serves its media through dedicated routes — one source of truth for code, prose, and assets.
- **Feature-detected rollouts.** Cross-repo integrations probe capabilities at runtime (inspect.signature, hasattr) so the two repositories can merge and deploy in any order without breaking each other.
- **Dependency injection at the seam.** kuantorflow injects its own save_flashcard into the agent's tool executor — the agent stays storage-agnostic; the site keeps one write path for all card sources.

- **LLM engineering in production:** TF-IDF RAG with citations, a streaming tool-use loop, persona/voice prompt engineering with few-shot examples, and defense-in-depth against prompt injection (names reduced to a single token, tool inputs sanitized to a whitelist).
- **Deterministic where determinism wins.** Farewell detection, date checks, and the goodnight reply are plain string/date logic — no model call, fully testable.
- **Security throughout:** parameterized SQL everywhere, XSS-safe DOM building for user-controlled strings, OAuth CSRF handled by Authlib, per-endpoint request caps, secrets in `.env` only, and consent copy kept honest as storage semantics changed.

Lessons Learned

1. **Wheel availability is the real Python-version constraint.** A "Visual C++ required" error almost always means a pinned package predates your interpreter (cpXYZ in the wheel name tells the story) — bump the pin, don't install a compiler. The same lesson set production's Python 3.13 floor via numpy/scipy.
2. **Same-name asset swaps fight browser caches.** Refreshing `background.jpg` and the favicon in place required hard refreshes; version query-strings are the durable fix.
3. **Keep disclosure in sync with behavior.** When per-user chat storage arrived, the consent popup's "nothing is stored on our server" had to change the same day — honesty is a feature requirement, not an afterthought.
4. **Verify with stubs, prove with one live call.** The test pattern that worked: exhaustive deterministic matrices with stubbed agents (free, fast), then a single real end-to-end call as the final proof — including cleanup of any real rows it created.
5. **Squash merges hide branch ancestry.** Post-merge cleanup needs `git branch -D` plus PR-state verification rather than trusting `-d`'s merged check.
6. **gitignore patterns don't recurse the way you hope.** `dir/chat_*.txt` does not cover `dir/sub/chat_*.txt` — user-identified logs nearly became committable the day subdirectories appeared.

Plans for Next Week

Statuses taken directly from the KuantorFlow Improvements GitHub Projects board (excluding this report's own ticket, kuantorflow#63, which the board lists as In Progress). The board also tracks 31 items as Done, matching the completed work summarized above.

In Progress

Repo	Issue	Title
kuantorflow	#62	Create DOCX tutorial documenting KuantorFlow development from scratch
kuantorflow	#54	Investigation: Android app clone of the KuantorFlow website
kuantorflow	#55	Make config.json unique for each Gmail user
kuantorflow	#46	Settings: add language visibility switches
kuantorflow	#45	UI: enhance behaviour of MHT-processing results window
kuantorflow	#30	Create logs for various actions in the app

kuantorflow	#21 / #20	Dictionary selection feature in Settings
kuantorflow	#13	Settings menu with auto-add cards option

Todo

Repo	Issue	Title
kuantorflow	#22	Redesign website for a modern look
kuantorflow	#56	Discussion: privacy disclosure & history bootstrap for per-user chat memory
kuantorflow	#25	Export/import cards (local ↔ remote database)
kuantorflow	#19	Tutorial tip above "Look Up & Save" button
kuantorflow	#15	Parallel PRs for features and tests
kuantorflow	#66	Create a separate board for done items
ai_agent	#19	Teach Mykola to complement and extend the meanings of cards
ai_agent	#47	Audit issue-61 changes: remove/refactor code not needed by the embedded widget
ai_agent	#28	UI: sync Mykola's description on both screens
kuantorflow_automation	#4	Basic automated tests for the AI agent

Report generated 13 July 2026 from GitHub PR and issue data across the three repositories (``gh pr list` / `gh issue list``), covering every merged PR from the first (kuantorflow#1, 9 July 2026) to the most recent (kuantorflow#64, 13 July 2026).